Assignment 8

Total = 10 Marks

Multiple Choice Questions

Question 1: Which of the following is a characteristic of non-parametric methods?

(a) They always assume a linear relationship
(b) They assume a fixed distributional form
(c) They do not assume any specific theoretical probability distribution of the data
(d) Often relies on a limited number of parameters

[1 *Mark*]

Correct answer: (c) They do not assume any specific theoretical probability distribution of the data.

Detailed Solution: Non-parametric methods do not assume any specific theoretical probability distribution of the data. The relationship is mainly determined from the observed data, which provides greater flexibility for capturing complex patterns.

Question 2: _____ occurs in machine learning when a model is too simple to capture the underlying structure or patterns in the data

(a) Underfitting
(b) Overfitting
(c) Optimized
(d) Parametric

[1 *Mark*]

Correct answer: (a) Underfitting

Detailed Solution: Underfitting occurs when a model is too simple to capture the underlying relationship between the predictors and the response. Such a model fails to represent important patterns in the data and generally produces relatively high errors on both training and testing datasets.

Question 3: Two broad caterogies of supervised learning is

(a) Clustering and Association
(b) Regression and Classification
(c) Regression and Prediction
(d) Sampling and Population

[1 *Mark*]

Correct answer: (b) Regression and Classification

Detailed Solution: Supervised learning uses input data together with known output values to learn the relationship between predictors and responses. It is mainly divided into regression, which predicts continuous numerical values, and classification, which predicts discrete categories.

Question 4: Reinforcement learning improves model performance mainly through:

(a) Increasing the number of predictors
(b) Imposing rewards and penalties
(c) Removing the outliers from the data
(d) None of these

[1 *Mark*]

Correct answer: (b) Imposing rewards and penalties

Detailed Solution: Reinforcement learning improves model performance through repeated interaction with an environment. Appropriate actions receive rewards, while undesirable actions incur penalties, enabling the learning process to identify actions that maximize overall reward.

Question 5: Pearson's correlation coefficient primarily measures

(a) Nonlinear association
(b) Linear association
(c) Categorical association
(d) None of these

[1 *Mark*]

Correct answer: (b) Linear association

Detailed Solution: Pearson's correlation coefficient measures the strength and direction of the linear relationship between two numerical variables. A value close to +1 indicates a strong positive linear association, while a value close to −1 indicates a strong negative linear association.

Question 6: Which value of the **Relative Absolute Error (RAE)** represents a perfect model?

(a) 0 (zero)
(b) 1
(c) $+\infty$
(d) None of these

[1 *Mark*]

Correct answer: (a) 0 (zero)

Detailed Solution: Relative Absolute Error (RAE) compares the total absolute prediction error with the error obtained from a reference model. An RAE value of zero indicates that the predicted values exactly match the observed values, representing a perfect model.

Question 7: An underfitted model generally has:

(a) Low bias and high variance
(b) High bias and low variance
(c) Low bias and low variance
(d) High bias and high variance

[1 *Mark*]

Correct answer: (b) High bias and low variance

Detailed Solution: An underfitted model is too simple to capture the underlying relationship present in the data. Such models generally produce high systematic error, resulting in high bias, while their predictions remain relatively stable across different training datasets, resulting in low variance.

Question 8: The optimal value of the ridge regularization parameter λ is generally estimated using:

(a) Trial and error
(b) Cross-validation
(c) Optimization
(d) None of these

[1 *Mark*]

Correct answer: (b) Cross-validation

Detailed Solution: Regularization techniques introduce a penalty on model complexity or large parameter values. This penalty restricts excessive model flexibility and reduces the model's tendency to fit random variations or noise in the training data. Therefore, regularization is mainly used to reduce overfitting.

Question 9: Occurrence/non-occurrence of cold wave prediction model produces the following outcomes in a 2x2 contingency table (Notations: TP - True Positive, FP - False Positive, TN - True Negative, and FN - False Negative):

Predicted ↓	**Observed**	
	Occurrence	**Non-Occurrence**
Occurrence	$TP = 40$	$FP = 10$
Non-Occurrence	$FN = 20$	$TN = 30$

What is the model accuracy?

(a) 0.70
(b) 0.60
(c) 0.50
(d) 0.40

[2 *Mark*]

Correct answer: (a) $\mathbf{0.70}$

Detailed Solution: Using the formula for accuracy,

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} = \frac{40 + 30}{40 + 10 + 30 + 20} = \frac{70}{100} = \mathbf{0.70}$$

Thus, the model correctly classifies 70% of the observations.